

Net

net accounts can be used to display the user accounts database and modify password and logon requirements for all accounts. Syntax: **net accounts** [/forcelogoff:{minutes | no}] [/minpwlen:length] [/maxpwage:{days | unlimited}] [/minpwage:days] [/uniquepw:number] [/domain]. Examples: 1) To display the current settings, the password requirements, and the server role for a server, type: *net accounts*. 2) To force users to log off after the logon time expires with a five-minute warning, type: *net accounts /forcelogoff:5*.

net config displays the configurable workstation and server services that are running. Syntax: **net config** [{server|workstation}]. To display computer name, DNS host name, logon user name, active NIC, logon domain, do *net config workstation*. To display server name, server active NIC, server hidden status and maximum logged on users, do *net config server*.

net file displays the names of all open shared files on a server and the number of file locks and closes individual shared files and removes file locks. Syntax: **net file** [ID [/close]]. To display the names and IDs of the open shared files, do *net file* without parameters. To close a file that has ID 5, type: *net file 5 /close*.

net send Sends messages to other users, computers, or messaging names on the network. Syntax: **net send** {name | * [/domain[:name] | /users]} message. To send "This server will shut down in 15 minutes" message to all users connected to the DOMAIN, type: *net send /domain:users This server will shut down in 15 minutes*.

net share displays information about all of the resources that are shared on the local computer. Syntax: **net share** [ShareName]

net start/stop starts/stop a service. Syntax: **net start/stop** [service]. To displays a list of services that are currently operating, use *net start* without parameters. To stop the Event Log service, type: *net stop eventlog*.

net statistics displays the statistics log for the local Workstation or Server service, errors, logon time and transfer bytes.. Syntax: **net statistics** [{workstation | server}]

net time display and synchronizes the computer's clock with that of another computer or domain. Syntax: **net time** [{\\ComputerName | /domain[:DomainName] | /rtdomain[:DomainName]]} [/set]. To synchronizes the computer's clock with the time on the specified computer or domain, type *net time \\computername /set*.

net use can be used to connect a computer to or disconnect a computer from a shared resource, or display information about computer connections. The command also controls persistent net connections. Syntax: **net use** [{DeviceName | *}] [\\ComputerName\ShareName[volume]] [{Password | *}]] [/user:[DomainName\UserName] [/user:[DottedDomainName\UserName] [/user: [UserName@DottedDomainName] [/savecred] [/smartcard] [{/delete | /persistent:{yes | no}}] To list network connections, type: *net use*. To connect to the resource accounting shared on the \\Financial data server, type: *net use k: "\\financial data" /accounting*.

net user adds or modifies user accounts or displays user account information such as user name, last logon, logon script, user profile, home directory and group memberships. Syntax: **net user** [UserName [Password | *] [options]] [/domain]. To view information about the user account jsmith in a domain, type: *net user jsmith /domain*. To specify logon hours of 4 A.M. until 5 P.M. on Monday, 1 P.M. until 3 P.M. on Tuesday, and 8 A.M. until 5 P.M. Wednesday through Friday for jsmith, type: *net user jsmith /time:M,4am-5pm;T,1pm-3pm;W-F,8:00-17:00*.

net view

net session

Netdom

Netdom is a command-line utility that you can use to manage Windows domains. It must be run from an elevated command prompt. Netdom uses the following general syntaxes:

```
NetDom <Operation> [<Computer>] [{/d: | /domain:} <Domain>] [<Options>]  
NetDom help <Operation>
```

```
netdom query {/d: | /domain:}<Domain> [{/s: | /server:}<Server>] [{/ud: | /userd:}[<Domain>\]<User> {/pd: |  
/passwordd}{<Password>|*}] [/verify] [/reset] [/direct] {WORKSTATION|SERVER|DC|OU|PDC|FSMO|TRUST}  
[{/help | /?}]
```

Netstat

Netstat displays protocol statistics and current TCP/IP connections. From a command prompt, type **Netstat -a** to display all connections and listening ports. Type **netstat -r** to display the contents of the IP routing table and any persistent routes. The **-n** switch tells Netstat not to convert addresses and port numbers to names, which speeds up execution. The **netstat -s** option shows all protocol statistics. The **netstat -p <protocol>** option can be used to show statistics for a specific protocol or together with the **-s** option to show connections only for the protocol specified. The **-e** switch displays interface statistics.

Nbtstat

Nbtstat is designed to help troubleshoot NetBIOS name resolution problems. When a network is functioning normally, NetBIOS over TCP/IP (NetBT) resolves NetBIOS names to IP addresses. It does this through several options for NetBIOS name resolution, including local cache lookup, WINS server query, broadcast, LMHOSTS lookup, Hosts lookup, and DNS server query.

The **nbtstat** command removes and corrects preloaded entries using a number of case-sensitive switches. The **nbtstat -a <name>** command performs a NetBIOS adapter status command on the computer name specified by **<name>**. The adapter status command returns the local NetBIOS name table for that computer as well as the MAC address of the adapter card. The **nbtstat -A <IP address>** command performs the same function using a target IP address rather than a name.

The **nbtstat -c** option shows the contents of the NetBIOS name cache, which contains NetBIOS name-to-IP address mappings.

nbtstat -n displays the names that have been registered locally on the system by NetBIOS applications such as the server and redirector.

The **nbtstat -r** command displays the count of all NetBIOS names resolved by broadcast and by querying a WINS server. The **nbtstat -R** command purges the name cache and reloads all #PRE entries from the LMHOSTS file. #PRE entries are the LMHOSTS name entries that are preloaded into the cache.

Nbtstat -RR sends name release packets to the WINS server and starts a refresh, thus re-registering all names with the name server without having to reboot. This is a new option in Windows NT 4.0 with Service Pack 4 as well as in Windows 2000.

You can use **nbtstat -S** to list the current NetBIOS sessions and their status, including statistics.

Netsh

NetSh is a network shell command line utility that allows you to display or even modify or otherwise change your Network Interface Card (NIC) settings or configuration on Microsoft windows XP or better local or remote Operating system machines.

With the NetSh utility you can also specify the command line arguments to run or execute a script so that you can execute multiple commands from the same execution with the -F switch followed by the script name.

NetSh Syntax And Command Switch Reference

The basic Syntax of the NetSh is shown directly below followed by a chart with the command line switch and a brief description of each.

NetSh [-a AliasFile] [-c Context] [-r RemoteComputer] [{NetshCommand|-f ScriptFile}]

Switch	Description
? or Help	Displays Help and provides this list of commands.
Add	Adds a new configuration entry to a list.
Bridge	Changes to the `NetSh Bridge' prompt context.
Delete	Deletes a configuration entry as opposed to Add entries.
Diag	Changes to the `NetSh Diag' prompt context.
Dump	Displays the contents of a configuration script.
Exec	Runs a script file.
Firewall	Changes to the `NetSh Firewall' prompt context.
Interface	Changes to the `NetSh Interface' prompt context.
Ras	Changes to the `NetSh Ras' prompt context.
Routing	Changes to the `NetSh Routing' prompt context.
Set	Updates configuration settings.
Show	Displays information available.
WinSock	Changes to the `NetSh WinSock' prompt context.

Netsh wireless network commands

```
netsh wlan
netsh wlan show networks
netsh wlan export profile name="<SSID>"
netsh wlan add profile filename="C:\profiles\nwtraders1.xml"
netsh wlan connect {NWtraders1}
netsh wlan disconnect
netsh wlan add filter permission=allow|denyall SSID= networktype={adhoc}
```

Event log IDs 11006, 8002, 12013 indicate wireless authentication failure

ARP

Arp allows you to view and modify the ARP cache. If two hosts on the same subnet cannot ping each other successfully, try running the **arp -a** command on each computer to see whether the computers have the correct media access control (MAC) addresses listed for each other. You can use Ipconfig to determine a host's correct MAC address.

You can also use Arp to view the contents of the ARP cache by typing **arp -a** at a command prompt. This displays a list of the ARP cache entries, including their MAC addresses.

You can use **arp -d < IP address >** to delete incorrect entries. Use **arp -s < MAC address >** (where the MAC address is formatted as hexadecimal bytes separated by dashes) to add new static entries; these static entries do not expire from the ARP cache. However, static entries do not persist after a reboot. For persistent static ARP cache entries, you must create a batch file run from the Startup group.

Use **arp -N < IP address >** to list all the ARP entries for the network interface specified by **< IP address >**.

Arp Switches

Switch	Name	Effect
-d < IP address >	Delete	Removes the listed entry from the ARP cache
-s < MAC address >	Static	Adds a static entry to the ARP cache
-N < Interface IP address >	Interface	Lists all ARP entries for the interface specified
-a	Display	Displays all the current ARP entries for all interfaces
-g	Display	Displays all the current ARP entries for all interfaces

IPconfig

IPConfig is a command-line tool that displays the current configuration of the installed IP stack on a networked computer.

Switch	Effect
/all	Produces a detailed configuration report for all interfaces.
/flushdns	Removes all entries from the DNS name cache.
/registerdns	Refreshes all DHCP leases and reregisters DNS names
/displaydns	Displays the contents of the DNS resolver cache.
/release < adapter >	Releases the IP address for a specified interface.
/renew < adapter >	Renews the IP address for a specified interface.
/showclassid < adapter >	Displays all the DHCP class IDs allowed for the adapter specified.
/setclassid < adapter > < classID to set >	Changes the DHCP class ID for the adapter specified.
/?	Displays this list.

Nslookup

Nslookup is a useful tool for troubleshooting DNS problems, such as host name resolution

Switch	Function
nslookup	Launches the nslookup program.
set debug	Launches debug mode from within nslookup.
set d2	Launches verbose debug mode from within nslookup.
host name	Returns the IP address for the specified host name.
NAME	Displays information about the host/domain NAME using default server
NAME1 NAME2	As above, but uses NAME2 as server
help or ?	Displays information about common commands
set OPTION	Sets an option
All	Displays options, current server and host.
[no]debug	Displays debugging information.
[no]defname	Appends domain name to each query.
[no]recurse	Asks for recursive answer to query.
[no]search	Uses domain search list.
[no]vc	Always uses a virtual circuit.
domain=NAME	Sets default domain name to NAME.
srchlist=N1[/N2/.../N6]	Sets domain to N1 and search list to N1,N2, and so on.
root =NAME	Sets root server to NAME.
retry=X	Sets number of retries to X.
timeout=X	Sets initial timeout interval to X seconds.
type=X	Sets query type (such as A, ANY, CNAME, MX, NS, PTR, SOA, SRV).
querytype=X	Same as type.
class=X	Sets query class (ex. IN (Internet), ANY).
[no]msxfr	Uses MS fast zone transfer.
ixfrver=X	Current version to use in IXFR transfer request.
Server NAME	Sets default server to NAME, using current default server.
Lserver NAME	Sets default server to NAME, using initial server.
Finger [USER	Fingers the optional NAME at the current default host.
Root	Sets current default server to the root.
ls [opt] DOMAIN [> FILE]	Lists addresses in DOMAIN (optional: output to FILE).
-a	Lists canonical names and aliases.
-d	Lists all records.
-t TYPE	Lists records of the given type (For example, A, CNAME, MX, NS, PTR and so on).
View FILE	Sorts the output file from the 'ls' option described earlier and displays it page by page.
Exit	Exits Nslookup and returns to the command prompt.

Pathping

The **pathping** command is a route tracing tool that combines features of the **ping** and **tracert** commands with additional information that neither of those tools provides. The default number of hops is 30, and the default wait time before a time-out is 3 seconds. The default period is 250 milliseconds, and the default number of queries to each router along the path is 100.

Switch	Name	Function
-n	Hostnames	Does not resolve addresses to host names.
-h	Maximum hops	Maximum number of hops to search for target.
-g	Host-list	Loose source route along host list.
-p	Period	Number of milliseconds to wait between pings.
-q	Num_queries	Number of queries per hop.
-w	Timeout	Waits this many milliseconds for each reply.
-T	Layer Two tag	Attaches a Layer Two priority tag (for example, for IEEE 802.1p) to the packets and sends it to each of the network devices in the path. This helps in identifying the network devices that do not have Layer Two priority configured properly. The -T switch is used to test for Quality of Service (QoS) connectivity.
-R	RSVP test	Checks to determine whether each router in the path supports the Resource Reservation Protocol (RSVP), which allows the host computer to reserve a certain amount of bandwidth for a data stream.

Tracert

Tracert is a route tracing utility that display a list of near-side router interfaces of the routers along the path between a source host and a destination. Tracert uses the IP TTL field in ICMP Echo Requests and ICMP Time Exceeded messages to determine the path from a source to a destination through an IP internetwork.

Switch	Function
-d	Specifies to not resolve addresses of router interfaces to host names.
-h < maximum_hops >	Specifies a maximum number of hops to reach destination.
-j < host_list >	Specifies loose source routing along the host-list.
-w < timeout >	Indicates how many milliseconds to wait for each reply.

DNScmd

This command-line tool assists administrators in Domain Name System (DNS) management.

Do any dnscmd command on a remote system	<i>dnscmd servername command</i>	<i>dnscmd main.bigfirm.com / zoneprint bigfirm.com</i>	
Create a primary zone	<i>dnscmd / zoneadd zonename /primary</i>	<i>dnscmd / zoneadd bigfirm.com /primary</i>	
Create a secondary zone	<i>dnscmd / zoneadd zonename /secondary master IP address</i>	<i>dnscmd / zoneadd bigfirm.com /secondary 192.168.1.1</i>	
Host a zone on a server based on an existing (perhaps restored) zone file	<i>dnscmd / zoneadd zonename /primary /file filename /load</i>	<i>dnscmd / zoneadd bigfirm.com /primary /file bigfirm.com.dns /load</i>	
Delete a zone from a server	<i>dnscmd / zonedeletem zonename [/f]</i>	<i>dnscmd / zonedeletem bigfirm.com /f</i>	(without the /f, dnscmd asks you if you really want to delete the zone)
Show all of the zones on a DNS server	<i>dnscmd / enumzones</i>	<i>dnscmd / enumzones</i>	
Dump (almost) all of the records in a zone	<i>dnscmd / zoneprint zonename</i>	<i>dnscmd / zoneprint bigfirm.com</i>	Doesn't show glue records.
Add an A record to a zone	<i>dnscmd / recordadd zonename hostname A ipaddress</i>	<i>dnscmd / recordadd bigfirm.com mypc A 192.168.1.33</i>	
Add an NS record to a zone	<i>dnscmd / recordadd zonename @ NS servername</i>	<i>dnscmd / recordadd bigfirm.com @ A dns3.bigfirm.com</i>	
Delegate a new child domain, naming its first DNS server	<i>dnscmd / recordadd zonename childname NS dnsservername</i>	<i>dnscmd / recordadd bigfirm.com test NS main.bigfirm.com</i>	This would create the "test.bigfirm.com" DNS child domain under the bigfirm.com DNS domain
Add an MX record to a zone	<i>dnscmd / recordadd zonename @ MX priority servername</i>	<i>dnscmd / recordadd bigfirm.com @ MX 10 mail.bigfirm.com</i>	
Add a PTR record to a reverse lookup zone	<i>dnscmd / recordadd zonename lowIP PTR FQDN</i>	<i>dnscmd / recordadd 1.168.192.in-addr.arpa 3 A pc1.bigfirm.com</i>	This is the PTR record for a system with IP address 192.168.1.3

Modify a zone's SOA record	<code>dnscmd / recordadd zonename @ SOA primaryDNSservername responsibleemailaddress serialnumber refreshinterval retryinterval expireinterval defaultTTL</code>	<code>dnscmd / recordadd bigfirm.com @ SOA winserver.bigfirm.com mark.bigfirm.com 41 1800 60 2592000 7200</code>	Ignores the serial number if it's not greater than the current serial number
Delete a resource record	<code>dnscmd / recorddelete zonename recordinfo [/f]</code>	<code>dnscmd / recorddelete bigfirm.com @ NS main.bigfirm.com /f</code>	Again, “/f” means “don’t annoy me with a confirmation request, just do it.”
Create a resource record and incorporate a nonstandard TTL	<code>dnscmd / recordadd zonename leftmostpartofrecord TTL restofrecord</code>	<code>dnscmd / recordadd bigfirm.com pc34 3200 A 192.168.1.4</code>	
Reload a zone from its zone file in \\windows\system32\ dns	<code>dnscmd / zonereload zonename</code>	<code>dnscmd / zonereload bigfirm.com</code>	Really only useful on primary DNS servers
Force DNS server to flush DNS data to zone file	<code>dnscmd / zonewriteback zonename</code>	<code>dnscmd / zonewriteback bigfirm.com</code>	
Tell a primary whom to allow zone transfers to	<code>dnscmd / zoneresetsecondaries zonename / nonsecure securens</code>	<code>dnscmd / zoneresetsecondaries bigfirm.com / nonsecure</code>	That example says to allow anyone who asks to get a zone transfer
Enable/disable DNS NOTIFY	<code>dnscmd / zoneresetsecondaries zonename /notify / nonotify</code>	<code>dnscmd / zoneresetsecondaries bigfirm.com / nonotify</code>	Example disables DNS notification, which is contrary to the default settings.
Tell a secondary DNS server to request any updates from the primary	<code>dnscmd / zonerefresh zonename</code>	<code>dnscmd / zonerefresh bigfirm.com</code>	
Enable or disable dynamic DNS on a zone	<code>dnscmd / config zonename / allowupdate 1 0</code>	1 enables, 0 disables, 0 is default	
Stop the DNS service	Either net stop dns or sc stop dns	(No dnscmd command for this)	
Start the DNS service	Either net start dns or sc start dns	(No dnscmd command for this)	

Install the DNS service on a 2008 full install system	servermanagercmd -install dns		
Install the DNS service on a 2008 Server Core system	ocsetup DNS-Server-Core-Role	Case matters — ocsetup dns -server-core-role would fail	
Uninstall the DNS service on a 2008 Server full install system	servermanagercmd -remove dns		
Uninstall the DNS service on a 2008 Server Core system	ocsetup /uninstall DNS-Server-Core-Role		